

MOHAMAD MOSTAFA

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PROFESSIONAL SUMMARY

Computer Science graduate with a decent to strong foundation in computer architecture, operating systems, networking, cybersecurity, software engineering, relational databases and full-stack web development. Currently expanding this range through a stream of self-driven engineering projects.

EDUCATION

Lebanese University — Faculty of Sciences, 3rd Branch

Bachelor of Science in Computer Science (B.Sc.) | GPA: 83/100

Graduation: 2026

TECHNICAL EXPERTISE

- **Languages:** C, C++, Rust, Assembly (MIPS32 / x86-64), Java, C#, Python, PHP, JavaScript (ES6+), PL/SQL, Prolog
- **Web & Apps:** React.js, Node.js / Express runtimes, C# ASP.NET WebForms, server-side PHP, HTML5, CSS3, mobile-first responsive design
- **Systems & Low-Level:** Process & thread management, Memory management, Kernel module development, System calls & interrupts, File systems & I/O, Concurrency & synchronization
- **Databases & Theory:** Oracle PL/SQL, SQL Server, ACID transactions, B+ Tree indexing, Compiler design, Formal Language Theory, Discrete Mathematics, Analysis, Linear & Abstract Algebra
- **Networking & Security:** CCNA Routing & Switching, CyberOps Fundamentals, Cisco Packet Tracer, Wireshark
- **Tools & Modeling:** Git/GitHub, Visual Paradigm (UML), ArcGIS Pro, Figma UI/UX, VS/VS Code, vim/neovim, Docker, Microsoft Office

PROFESSIONAL EXPERIENCE

Junior Independent Computer Science Teacher

2023 —Present

Self-employed / Mentorship

- Instructed university peers and early-stage students in advanced data structures, systems programming in C, programming paradigms, structural networking fundamentals and various other subjects.

CORE SOFT SKILLS & LANGUAGES

- **Core Competencies:** Collaboration, communication, project management, agility, emotional intelligence, learning velocity and critical thinking.
- **Languages:** Arabic (Native), English (Fluent), French (Good).

SELECTED ENGINEERING PROJECTS

MediCare Healthcare Platform [C#, ASP.NET WebForms, SQL Server]

A full-stack, multi-role healthcare administration platform (Patient, Doctor, Admin). Implements BCrypt-secured authentication, role-based master pages, real-time chat, slot-based appointment scheduling, and a notification system backed by transactional SQL Server data flows.

Repository: github.com/moudzx/MediCare

Hybrid Parallel Distributed File Search [C, MPI, OpenMP]

A high-throughput text search engine combining distributed and shared-memory parallelism. Uses MPI to coordinate cross-node workload distribution and pthreads/OpenMP for concurrent in-node searching, with iterative optimization of scatter, broadcast, and point-to-point communication.

Repository: github.com/moudzx/Parallel-programming

CPU Scheduling Real-Time Visualizer [C, Raylib]

An interactive graphical simulator built with Raylib that animates Gantt charts for FCFS, SJF, Round Robin, and Priority scheduling, including both preemptive and non-preemptive variants, with full-window scrolling.

Repository: github.com/moudzx/CPU-scheduling-visualizer

Helmert 7-Parameter Geodetic Transformer [Java Swing, C]

A geodetic coordinate transformation tool implemented in both a Java Swing desktop application and a C terminal utility, performing seven-parameter Helmert transformations between geographic and geocentric coordinate systems.

Repository: github.com/moudzx/Helmert-transformation

Linux Loadable Kernel Module [C, Linux Kernel Internals]

A low-level loadable kernel module demonstrating manual initialization hooks, ring-buffer logging, hardware configuration tracking, and safe memory-mapped cleanup on unload.

Repository: github.com/moudzx/Linux-driver

Football League Database Engine (RDBMS) [Oracle SQL, PL/SQL]

A relational database system with custom PL/SQL packages enforcing match validation rules, B+ Tree-backed indexing, and strict ACID transactional integrity.

Repository: github.com/moudzx/RDBMS

Theory of Computation Educational Toolkit [Automata Theory]

An algorithmic suite for abstract computing models: interactive DFA design, automated ϵ -NFA conversion, regular expression parsing, language operations, context-free grammars, and Pushdown Automata simulation.

Repository: github.com/moudzx/TheoryOfComputation

MIPS32 Cryptographic Memory Maze [MIPS32 Assembly, C]

A hybrid bare-metal utility combining MIPS32 assembly with C to manipulate volatile memory matrices, tracking precise register maps, branch offsets, and byte-level transformations under tight hardware constraints.

Repository: github.com/moudzx/MIPS32-Assembly

Prolog Heuristic State-Space Search Engine [Prolog, Classical AI]

A declarative logic framework comparing DFS, BFS, IDS, and A* search on grid pathfinding, including a 6x6 grid with dead-end traps to demonstrate A*'s heuristic advantage.

Repository: github.com/moudzx/Prolog

Cinema Booking System Architecture [UML, Visual Paradigm, OOP]

An object-oriented design blueprint for a concurrent ticket-reservation engine, developed in Visual Paradigm with a full set of Class, Sequence, Use Case, and Activity diagrams.

Repository: github.com/moudzx/architectural-design

Meow — POSIX Interrupt Signal Controller [C, POSIX Signals]

A low-level utility that intercepts native Linux terminal hooks, overriding SIGINT (Ctrl+C) via a custom POSIX signal handler that links live terminal interrupts to local audio playback.

Repository: github.com/moudzx/Meow

Letterboxd Activity Telemetry Tracker [Python, SQLite, BeautifulSoup]

An unfollower-tracking automation tool with both a client-side web app and a Python CLI (watch/check/history/clear), using SQLite snapshots and scheduled scraping to monitor follower changes.

Repository: github.com/moudzx/Letterboxd-tracker

Tripoli Culture Portal [HTML5, CSS3, JavaScript, RTL Arabic]

A responsive static web interface showcasing the history and landmarks of Tripoli, Lebanon, featuring glassmorphism UI panels, smooth scroll physics, and full Arabic (RTL) localization.

Repository: github.com/moudzx/static-web

PHP Software Suite [PHP, Relational Databases]

A modular collection of web applications, including a COVID tracker, a geographical borders system, a library management engine, and standalone product, student course, and attendance-tracking tools.

Repository: github.com/moudzx/PHP

Dark Shadows [HTML5 Canvas, JavaScript]

A 2D arcade game made with JavaScript, featuring custom particle physics, linear and radial camera tracking interpolation, and a dynamic rhythmic pacing system that alternates intensities using sine-wave frequency modifiers.

Repository: github.com/moudzx/DarkShadows